RAKHESH NAMINENI

Junior AI Engineer | Python | Generative AI & LLMs | Machine Learning | RAG | REST APIs
9392475264 | namenirakesh@gmail.com | github.com/Rakhesh143 | linkedin.com/in/rakhesh-namineni431 |
huggingface.co/Namineni1

PROFESSIONAL SUMMARY

Recently graduated AI engineer with a year of hands-on experience building practical AI applications in Python. Experienced across the workflow: data cleaning, preprocessing and validation, model fine-tuning and evaluation, and automated testing of AI outputs. Built LLM and RAG applications (prompt engineering, embeddings, vector search with LangChain and LangGraph) and connected them to web applications through REST APIs, FastAPI services, TypeScript and PostgreSQL. Validated a multilingual support chatbot across 205 automated test categories at 98% accuracy before release. Eager to keep learning, collaborate with engineers and apply AI to real business problems.

TECHNICAL SKILLS

Programming & Data: Python (NumPy, Pandas), SQL, JavaScript, TypeScript, JSON; data cleaning, preprocessing and validation, data analysis, semantic chunking, embeddings

AI / ML: scikit-learn, PyTorch, TensorFlow, Hugging Face; model fine-tuning (SFT), model evaluation, custom eval harnesses, multi-category automated test suites (205 categories)

Generative AI & LLMs: Prompt engineering, RAG, LangChain, LangGraph (multi-agent), tool/function calling, memory systems; LLM APIs: OpenAI, Anthropic, Gemini, Groq, Hugging Face, AWS Bedrock

Web & APIs: HTML, CSS, JavaScript, React (used in CARE chatbot project), REST API design, FastAPI, Supabase Edge Functions, microservices, GraphQL (familiar)

Databases & Vector Stores: PostgreSQL (incl. PGVector, row-level security), MySQL, FAISS, ChromaDB, Pinecone (familiar)

Cloud & Tools: AWS (SageMaker, EC2, S3, Bedrock), Docker, Git (feature branching, PR review), CI/CD (GitHub Actions), pytest, Railway, Modal.com, Supabase

WORK EXPERIENCE

Cognysis AI | Hyderabad, India — Jul 2025 - Jul 2026
AI Engineer - Generative AI & NLP

- Built a multilingual (English, Spanish, Telugu) RAG-based customer-support chatbot and deployed it on client websites, with human-in-the-loop escalation for low-confidence queries.
- Validated the chatbot with 205 automated test categories (security, typos, multilingual, edge cases), reaching 98% accuracy before release.
- Owned pretraining data pipelines for LUCA V3, a production Telugu LLM (data cleaning, preprocessing and quality analysis) and SFT dataset curation, using synthetic data from Gemma 4 26B on Modal.com and large-scale Telugu transcript extraction from YouTube with yt-dlp.
- Designed vendor-agnostic RAG with FAISS vector search, embeddings and semantic chunking; ~85% accuracy improvement over keyword search on 500+ page corpora.
- Developed a reusable LLM provider abstraction in modular Python: OpenAI, Gemini and Groq behind one common interface, so RAG code can switch providers without code changes.
- Fine-tuned and served LLMs on AWS SageMaker with EC2 inference endpoints, cutting processing time by ~60%.
- Trained and benchmarked tokenizers for 10 Indic languages (fertility, compression efficiency) on SageMaker and published them to Hugging Face Hub (see Akshara project).
- Evaluated paid Telugu dataset vendors and drafted AWS H100/A100 GPU quota requests for model training.

PROJECTS

Enterprise Multilingual CARE Support Chatbot & Automation Engine

Supabase Edge Functions (TypeScript), Gemini 3 Flash, OpenAI Embeddings, PostgreSQL (PGVector), React

- **Architecture:** modular, decoupled from any single LLM vendor; Edge Functions orchestrate Gemini 3 Flash and OpenAI embeddings behind a swappable interface, with a 2-stage RAG pipeline (PGVector search + LLM context filtering).
- **Evaluation:** 98% accuracy across 205 automated test categories (Security, Typos, Multilingual, Edge Cases), used to catch regressions before release.
- **Guardrails and integrations:** automated human-in-the-loop ticket escalation for low-confidence queries (<0.7 threshold), PostgreSQL RLS access control and database-driven config toggles for auditability and oversight.
- **Cost and latency by design:** native cross-lingual handling (English, Spanish, Telugu) with no external translation API, keeping average response latency at 1.2s.

RAG-Based PDF Question Answering System

Python, LangChain, FAISS, OpenAI API, Vector DB, Docker

- Built a modular RAG application for natural-language Q&A over large enterprise PDFs: semantic chunking, FAISS vector search, embedding generation and LLM answers via the OpenAI API.
- Containerized the application with Docker during deployment to keep environments consistent and eliminate dependency and environment-mismatch conflicts.

Akshara Multilingual Tokenizer

Hugging Face, AWS SageMaker, Python, SentencePiece/BPE, FastAPI

- Built and trained tokenizers for 10 Indic languages on AWS SageMaker, tuning vocabulary size using fertility and compression benchmarks; published to Hugging Face Hub with a FastAPI demo microservice for validation.

EDUCATION

B.Tech - Computer Science & Engineering — 2022 - 2026
QIS College of Engineering and Technology, Ongole | CGPA: 7.7 / 10

Intermediate (MPC) — 2020 - 2022
Sri Chaitanya Jr. College, Ongole | 904 / 1000

SSC — 2020
Nirmala High School, Ongole | GPA: 10.0 / 10

CERTIFICATIONS & ACHIEVEMENTS

- Oxlo.ai Hackathon - Top 15 Finalist (HireSense AI Hiring Platform)
- Led a cross-functional AI project team to on-time delivery of production AI systems
- Presented and demonstrated AI/ML projects, explaining technical concepts and project functionality clearly while collaborating with team members
- Python & SQL - HackerRank | Generative AI for Beginners - Great Learning
- 2nd Prize - Essay Writing Competition, College Level

Languages: English (Fluent), Hindi, Telugu (Native)